8051 MICROCONTROLLER ARCHITECTURE

Exam-ready notes: simplified, shortened, with comments. Source: Module 3 Parts 1, 2, 3 (slides). [Ref] = standard 8051 fact not in the slides; the slides' errors are listed in section 14.

1. INTRODUCTION: MICROCONTROLLER vs MICROPROCESSOR

- **Microcontroller (MC)**: CPU + RAM + ROM + I/O + timer + serial port **on one chip**. Fixed amount of on-chip memory/ports. Used where cost, power and space matter (single purpose).
- **Microprocessor (MP)**: CPU only (x86, 680x0). RAM, ROM, I/O and timers are **external**. General purpose, expandable, versatile.
- **History**: Intel 8048 (1976: 1K EPROM, 64B RAM, 27 I/O). **Intel 8051 (1980)**: 4K ROM, 128B RAM, 2 x 16-bit timers, serial port, 32 I/O - the most popular 8-bit MC. Others: 8096 (16-bit, 1982), PIC16C64 (1985), ARM (32-bit, high-end).
- **Architecture types**: **Von Neumann** = one memory and one bus for code and data. **Harvard** = separate program and data memory/buses. The 8051 has separate code and data spaces (Harvard style).

MCS-51 family

Feature	8031	8051	8052	8751
On-chip ROM	None	4K	8K	4K EPROM
RAM (bytes)	128	128	256	128
Timers	2	2	3	2
I/O pins	32	32	32	32
Serial port	1	1	1	1
Interrupt sources	6	6	8	6

EXAM TIP: the 8051 = 4K ROM, 128B RAM, 2 timers, 4 ports (32 I/O), 1 serial port. Slide table lists 6 interrupt sources for the 8051; the standard count is 5 (INT0, T0, INT1, T1, serial). Confirm with your teacher.

2. BLOCK DIAGRAM & KEY FEATURES

```
XTAL1/XTAL2 -> [Oscillator & timing]
+-------------------------------------------------+
|  8-bit CPU: ALU, A, B, PSW, SP, DPTR, PC        |
|  Program ROM 4K      Data RAM 128B (+SFRs)      |
|  Timer0, Timer1 (16-bit)   Serial UART (SBUF)   |
|  Interrupt control (5 sources)                  |
|  P0  P1  P2  P3  -> 32 I/O lines                |
+-------------------------------------------------+
  internal data bus connects all of the above
```

- 8-bit CPU; 4K ROM; 128B RAM; 4 x 8-bit ports (32 I/O); 2 x 16-bit timer/counters; full-duplex serial UART; 5 interrupts; 64K code + 64K external data space; bit-addressable (Boolean) processor; 111 instructions.
- **Machine cycle (MC)** = 12 oscillator clocks (fetch/decode/execute). Some chips differ (Dallas: 1 clock per MC). [Ref] standard 8051 = 12 clocks/MC.

3. PIN DESCRIPTION (40 pins; 32 I/O; 24 dual-purpose)

Pin	Name	Function / comment
29	PSEN	Program Store Enable (output). 0 = read external **code** memory (OE of ROM).
30	ALE / PROG	Address Latch Enable: separates the multiplexed low address from data on P0. PROG = flash programming pulse.
31	EA / VPP	External Access. **EA = 0** -> code fetched from external memory only; **EA = 1** -> internal ROM first. VPP = 12 V for flash programming.
9	RST	Reset input (active high). [Ref] PC = 0000H after reset.
18, 19	XTAL1, XTAL2	Crystal / oscillator input and output. [Ref] Clock = 11.0592 MHz in many examples.
32-39	P0.0-P0.7	Port 0: **open-drain** -> needs 10 kΩ pull-up for I/O. Multiplexed low address/data bus when external memory is used.
1-8	P1.0-P1.7	Port 1: general I/O. Internal pull-up; no external resistors.
21-28	P2.0-P2.7	Port 2: general I/O, or **high address byte** with external memory.
10-17	P3.0-P3.7	Port 3: dual function. [Ref] P3.0 RXD, P3.1 TXD, P3.2 INT0, P3.3 INT1, P3.4 T0, P3.5 T1, P3.6 WR, P3.7 RD.

EXAM TIP: Port 3 pins have alternate functions; avoid using them as plain I/O. Port 0 needs external pull-ups; P1/P2/P3 do not.

4. MEMORY ORGANIZATION

- **Program memory (ROM/code)**: up to 64 KB (16-bit address). Internal 4K + external. Read with MOVC and fetches (PSEN).
- **Data memory**: internal RAM (128 B; 256 B on 8052) + external RAM up to 64 KB. External RAM is accessed only with MOVX (RD/WR).
- Same address can mean different things in code vs data space; the instruction tells the CPU which one (Harvard idea).

Internal RAM map (00H-7FH)

Address	Use
00H-1FH	**4 register banks** x 8 registers (R0-R7). Bank 0 is selected after reset.
20H-2FH	**Bit-addressable** RAM: bytes 20H-2FH hold bits **00H-7FH**. e.g. bit 00H = bit0 of 20H; bit 7FH = bit7 of 2FH.
30H-7FH	General-purpose RAM (scratchpad). Stack normally starts here.
80H-FFH	**SFR space** (direct addressing only). Ports, timers, PSW, etc.

- **Register bank** chosen by PSW bits RS1 RS0: 00 = bank 0 (00-07H), 01 = bank 1 (08-0FH), 10 = bank 2 (10-17H), 11 = bank 3 (18-1FH). Address of Rn = 8 x bank + n (R3 in bank 2 = 13H).
- Bit-addressable SFRs have addresses ending in 0 or 8: 80, 88, 90, 98, A0, A8, B0, B8, D0, E0, F0.

5. REGISTERS & SFRs

- **A (ACC, E0H)**: accumulator; used by almost all ALU operations.
- **B (F0H)**: used with A for MUL and DIV (also a general register).
- **PC**: 16-bit program counter (0000H after reset). **DPTR** (DPH 83H + DPL 82H): 16-bit data pointer.
- **SP (81H)**: 8-bit stack pointer, reset value 07H.

Special Function Registers (SFRs)

SFR	Addr	Function	SFR	Addr	Function
ACC	E0	Accumulator	TCON	88	Timer control (flags, run)
B	F0	Mul/Div	TMOD	89	Timer mode
PSW	D0	Program status flags	TL0/TH0	8A/8C	Timer 0 low/high byte
SP	81	Stack pointer	TL1/TH1	8B/8D	Timer 1 low/high byte
DPL/DPH	82/83	Data pointer low/high	IE	A8	Interrupt enable
P0/P1	80/90	Port latches 0/1	IP	B8	Interrupt priority
P2/P3	A0/B0	Port latches 2/3	SCON	98	Serial control
PCON	87	Power control (SMOD)	SBUF	99	Serial data buffer

PSW - Program Status Word (D0H)

Bit	7	6	5	4	3	2	1	0
Name	CY	AC	F0	RS1	RS0	OV	-	P

- **CY** carry (bit 7 out / borrow); **AC** carry out of bit 3 (BCD); **F0** user flag; **RS1/RS0** bank select; **OV** signed overflow; **P** parity = 1 if A has an odd number of 1s.

EXAM TIP: SFRs are reached by name or address. Since 80H-FFH is SFR space, only direct addressing reaches them (no @Ri). Reset: ports = FFH, SP = 07H, PC = 0000H, most others = 00H.

6. STACK

- Stack lives in internal RAM. **SP** points to the last byte used. Default SP = 07H, so the stack starts at 08H.
- **PUSH**: SP = SP + 1, then copy byte to [SP]. **POP**: copy [SP] to byte, then SP = SP - 1.
- **ACALL / LCALL** push the return PC: low byte first, then high byte (SP += 2). **RET / RETI** pop PC (SP -= 2).

EXAM TIP: the 8051 stack grows **UP** (toward higher addresses). Before using register bank 1-3 or a big stack, move SP above the banks (e.g. MOV SP,#2FH) so data is not overwritten.

7. I/O PORTS

- 4 bidirectional 8-bit ports (P0-P3). Each pin is bit-addressable (e.g. SETB P1.2). Reset: all ports = FFH (output high).
- To use a port as **input**, first write 1 to it: `MOV P1,#0FFH ; MOV A,P1`.
- **P0**: open-drain, no internal pull-up; a 1 floats the pin, so a 10k pull-up is needed for output-high. **P1-P3**: internal weak pull-ups.
- Read-modify-write instructions (ANL, ORL, CPL, INC, DJNZ, SETB/CLR bit) read the **latch**, not the pin. [Ref]

```
MOV A,#22       ; A = 22H
MOV P0,A        ; send to port 0 (needs pull-ups)
ACALL DELAY     ; wait
CPL A           ; complement A
SJMP BACK       ; repeat forever
```

8. ADDRESSING MODES

Mode	Example	Meaning / range
Register	MOV R0,A	Operand is in Rn of the current bank.
Direct	MOV 30H,A ; MOV A,P3	Operand = RAM or SFR address (or SFR name).
Indirect	MOV @R0,A ; MOVX A,@DPTR	Address held in R0/R1 (8-bit) or DPTR (16-bit). Only R0, R1, DPTR (and SP for stack).
Immediate	ADD A,#80H ; MOV DPTR,#0FE00H	Constant in the instruction; '#' marks data.
Relative	SJMP AHEAD ; JNZ LOOP	8-bit signed offset: **-128 to +127** bytes from next instruction.
Absolute	AJMP / ACALL addr11	11-bit address; target must be in the **same 2 KB page** as the next instruction.
Long	LJMP / LCALL addr16	16-bit address; anywhere in 64 KB code space.
Indexed	MOVC A,@A+DPTR ; MOVC A,@A+PC	Look-up table in code memory: A + base (DPTR or PC).

EXAM TIP: Relative/Absolute/Long apply to jumps and calls. Opcode sizes: AJMP/ACALL/SJMP = 2 bytes; LJMP/LCALL = 3 bytes.

9. INSTRUCTION SET (opcode + operands; format: OP Destination, Source)

Notation: Rn = R0-R7 (current bank); @Ri = R0/R1 indirect; direct = RAM/SFR address; bit = bit address; rel = signed 8-bit offset; #data = immediate.

A. Arithmetic

- **ADD A,src**: A = A + src. **ADDC A,src**: A = A + src + CY. Flags: CY, AC, OV.
- **SUBB A,src**: A = A - src - CY (no plain SUB on 8051; use CLR C first). Flags: CY, AC, OV.
- **INC / DEC** byte (no flags). **INC DPTR**: only 16-bit increment. DEC 00H underflows to FFH.

- **MUL AB**: B:A = A x B (B = high byte, A = low byte). OV = 1 if result > FFH; CY = 0. 4 cycles. Example: 85 x 23 = 1955 = 07A3H -> B = 07H, A = A3H, OV = 1.
- **DIV AB**: A = quotient, B = remainder. CY = 0; OV = 1 if B was 0 (result undefined). Example: 90 / 5 -> A = 12H, B = 00H.
- **DA A**: decimal adjust after ADD/ADDC on packed BCD (adds 06H / 60H).

B. Logical (no flags unless noted)

- **ANL / ORL / XRL** dst,src = AND / OR / XOR bit by bit. Used for masking: ANL P1,#10111001B clears bits 6,2,1.
- **CLR A** (A = 0), **CPL A** (complement), **RL / RR A** (rotate 1 bit), **RLC / RRC A** (rotate through CY), **SWAP A** (swap nibbles). Example: SWAP on C3H -> 3CH.

C. Data transfer

- **MOV dst,src**: copies byte; source unchanged. **MOV DPTR,#data16** loads 16-bit pointer (DPH:DPL).
- **MOVX**: access **external** RAM (A @Ri or @DPTR). **MOVC**: read **code** ROM table (A @A+DPTR or @A+PC).
- **PUSH / POP direct**: stack operations. **XCH A,src**: swap A with byte. **XCHD A,@Ri**: swap low nibbles only.

D. Boolean (bit) instructions

- **CLR / SETB / CPL** bit or C. **ANL / ORL C,bit** (and **/bit** = use inverted bit, source unchanged). **MOV C,bit / MOV bit,C**. Bit addresses: P1.2 or 92H; ACC.7 = E7H.
- **JC / JNC rel** (test CY), **JB / JNB bit,rel** (test bit), **JBC bit,rel** (test, clear bit, jump).

E. Program branching

- **SJMP rel**, **AJMP addr11**, **LJMP addr16**, **JMP @A+DPTR** (computed jump, e.g. jump table).
- **ACALL / LCALL** call subroutine, **RET** return. **RETI** returns from ISR and re-enables interrupts; PSW is not restored automatically.
- **JZ / JNZ rel**: branch if A = 0 / A != 0. **CJNE dst,src,rel**: jump if not equal; CY = 1 if dst < src (unsigned).
- **DJNZ Rn/direct,rel**: decrement, jump if not zero (loop counter). **NOP**: 1 cycle, used for timing.

EXAM TIP: flags: CY is changed by ADD/ADDC/SUBB, MUL/DIV (=0), RLC/RRC, CLR/SETB/CPL C, bit ops with C. AC: ADD/ADDC/SUBB only. OV: ADD/ADDC/SUBB, MUL, DIV. P: updated by any instruction that changes A. Rotates, logic and MOV are otherwise flag-free.

```
MOV R1,#60H     ; pointer R1 = 60H
MOV A,@R1       ; A = [60H]      (indirect)
MOV R2,#61H
ADD A,@R2       ; A = A + [61H]  (flags set)
MOV R7,A        ; R7 = A
; if [60H]=10H and [61H]=20H  ->  R7 = A = 30H
```

10. MACHINE CYCLES & TIME DELAYS

- **Machine cycle** = 12 clocks. MC time = 12 / f(XTAL). 11.0592 MHz -> MC = 1.085 µs. 16 MHz -> 0.75 µs. 20 MHz -> 0.6 µs.

Cycles	Instructions (examples)
1	Most: MOV Rn,#data; MOV A,Rn; INC/DEC; ADD; CLR/CPL; SETB; ANL/ORL with A or Rn; NOP; RR/RL; SWAP
2	MOVX, MOVC, PUSH, POP, ACALL, LCALL, AJMP, LJMP, SJMP, RET, RETI, DJNZ, CJNE, JZ, JNZ, JB, JNB, JC, JNC, MOV DPTR,#16
4	MUL AB, DIV AB

- **Delay = (number of cycles executed) x MC time.** A DJNZ loop costs 2 cycles per pass. Example: MOV R3,#200 (1) + [DJNZ R3 (2)] x 200 + RET (2) = 403 cycles -> 403 x 1.085 = 437.255 µs.
- Nested loops multiply the counts (e.g. R2 outer x R3 inner); the outer loop's setup instruction adds a little extra time.

EXAM TIP: delay formula: count x (cycles per loop) x MC. Counting the setup/RET instructions makes it exact; usually stated as approximate.

```
DELAY: MOV R3,#200   ; 1 cycle
HERE:  DJNZ R3,HERE  ; 2 cycles per pass, 200 passes
       RET           ; 2 cycles
; total = (200*2 + 1 + 2) * 1.085 us = 437.255 us
```

11. TIMERS / COUNTERS

- **Timer 0 and Timer 1**: 16-bit each (TH:TL). Count up each machine cycle (timer mode) or on falling edges of pins T0 (P3.4) / T1 (P3.5) (counter mode). Each is read/written as two 8-bit registers (TL0/TH0, TL1/TH1).
- **TMOD (89H)** (not bit-addressable): **upper nibble = Timer 1**, **lower nibble = Timer 0**. Each nibble: **GATE** (bit 3/7), **C/T** (bit 2/6: 0 = timer, 1 = counter), **M1 M0** (bits 1-0 / 5-4 = mode).
- **TCON (88H)** (bit-addressable): **TF1** (7) timer 1 overflow flag; **TR1** (6) run timer 1; **TF0** (5) timer 0 overflow; **TR0** (4) run timer 0; IE1 (3), IT1 (2), IE0 (1), IT0 (0) = interrupt flags/types. [Ref]
- **Start/stop**: SETB TRx starts, CLR TRx stops (when GATE = 0). With GATE = 1, the timer runs only while INTx pin (P3.2 for T0, P3.3 for T1) is high AND TRx is set.

Timer modes (M1 M0)

M1 M0	Mode	Description
00	0 - 13-bit	TH (8 bits) + TL (lower 5 bits = prescaler). Max count 8192. Mostly for old MCS-48 compatibility.
01	1 - 16-bit	TH:TL 16 bits, count to FFFFH then roll to 0000H, set TF. **No auto-reload**: reload TH/TL in software every time.
10	2 - 8-bit auto-reload	TL counts; on overflow TF is set and TL is reloaded from TH. Good for baud rates and fixed-rate tones.
11	3 - split	Timer 0 = two 8-bit timers (TL0 uses TR0/TF0; TH0 uses TR1/TF1). Timer 1 holds its count (stopped).

Calculations (XTAL = 11.0592 MHz, MC = 1.085 µs)

- **Mode 1**: count = delay / MC; load value = 65536 - count (TH = high byte, TL = low byte). Example 5 ms: 5000 / 1.085 ≈ 4608; 65536 - 4608 = 60928 = **EE00H** -> TH = EEH, TL = 00H.
- **Mode 2**: TH = 256 - count (e.g. TH = 5 -> (256-5) x 1.085 µs = 272.3 µs high time; period = 544.7 µs; f ≈ 1.836 kHz).
- **Mode 0**: load value = 8192 - count (13-bit count range 0-8191).

Mode 1 program steps (standard loop)

- 1) Load TMOD 2) Load TL and TH 3) SETB TRx (start) 4) JNB TFx,target (wait for overflow) 5) CLR TRx (stop) 6) CLR TFx (clear flag) 7) reload TL/TH, go back to step 3.

```
MOV TMOD,#01H   ; timer 0, mode 1 (16-bit)
MOV TL0,#00H    ; low byte  (EE00H = 60928 = 65536 - 4608)
MOV TH0,#0EEH   ; high byte -> 5 ms
SETB TR0        ; start timer 0
WAIT: JNB TF0,WAIT ; wait for overflow
CLR TR0         ; stop
CLR TF0         ; clear flag for next round
```

Mode 2 program steps (auto-reload)

- 1) Load TMOD 2) Load TH (only once) 3) SETB TRx 4) JNB TFx 5) CLR TFx (no reload needed) 6) go to step 4.

EXAM TIP: Counter: C/T = 1; counts external pulses on T0 (P3.4) or T1 (P3.5). Set the pin high first (SETB P3.5). Example: MOV TMOD,#01100000B = counter 1, mode 2. [Ref] counts falling edges; max input rate = fosc/24.

12. INTERRUPTS (not in the slides' content - standard 8051 reference)

- Interrupt sources: **INT0** (P3.2), **Timer 0 (TF0)**, **INT1** (P3.3), **Timer 1 (TF1)**, **Serial (RI/TI)**.
- **IE (A8H)**: **EA** bit 7 = global enable; ES (4), ET1 (3), EX1 (2), ET0 (1), EX0 (0). Set EA = 1 and the source bit = 1 to enable.
- **IP (B8H)**: 1 = high priority for each source (PS, PT1, PX1, PT0, PX0). A high-priority interrupt cannot be interrupted by a low one; equal priority is resolved by a fixed order.

Source	Vector (ISR address)
External 0 (INT0)	0003H
Timer 0 (TF0)	000BH
External 1 (INT1)	0013H
Timer 1 (TF1)	001BH
Serial (RI or TI)	0023H

- On an interrupt: CPU finishes the current instruction, pushes PC (LCALL-style), jumps to the vector. ISR ends with **RETI**. Timer flags are cleared by hardware when the vector is taken; INT0/INT1 are falling-edge triggered when IT=1, low-level triggered when IT=0.

EXAM TIP: ISR vectors are only 8 bytes apart, so put a jump (LJMP) at the vector to reach a longer routine.

13. SERIAL PORT (brief)

- Full-duplex UART: TXD (P3.1) and RXD (P3.0). SBUF (99H) = transmit/receive buffer (write = send, read = received byte).
- **SCON (98H)**: SM0 SM1 select mode; REN (4) enable receive; TI (1) transmit done; RI (0) receive done (set by hardware, cleared by software).
- Modes: 0 = shift register (fosc/12); 1 = 8-bit UART (baud from Timer 1 mode 2); 2 = fixed fosc/64; 3 = 9-bit UART (variable baud).

14. QUICK FORMULAS & SLIDE CORRECTIONS

Quantity	Formula / value
Machine cycle	MC = 12 / fXTAL (11.0592 MHz -> 1.085 µs)
Timer 1 mode 1 load	65536 - (delay / MC)
Timer mode 2 load	256 - (delay / MC) (TH:TL auto reload)
Square-wave frequency	f = 1 / (2 x half-period); half-period = (count x MC)
Loop delay	count x (cycles in loop) x MC
Bit address 20H-2FH	00H-7FH

Errors found in the source slides (check before the exam)

- **Example 3-15** slide: total shown as 436.255 µs; the correct sum is 403 x 1.085 = **437.255 µs**.
- **SUBB example** slide: A = C1H, R1 = 40H, CY = 1 gives **80H**, CY = 0, AC = 0, OV = 0 (slide shows 70H, OV = 1).
- **Interrupt count**: slide table says 6 sources for 8051; standard count is 5.

Last-minute checklist

- MC vs MP; 8051 = 4K ROM, 128B RAM, 2 timers, 32 I/O, serial. Harvard. EA, PSEN, ALE roles. P0 open drain.
- Register banks (00-1F), bit area (20-2F), SFR (80-FF). PSW flags CY AC OV P. Stack grows up; SP default 07H.
- Addressing modes; jump ranges (SJMP ±128, AJMP 2K page, LJMP 64K); MUL/DIV results in A and B.
- Timers: TMOD nibbles, TCON run/flag bits, modes 0-3, load formulas, counter mode on T0/T1.